Lie Theory: Chapter 5

Connor Mooneyhan

1. Let V be a vector space, F be a field, and A be a subalgebra of $\mathfrak{gl}(V)$. Give an example of a function $f : A \to F$ whose eigenspace is zero.

Proof. Define f to be the map sending every element to 1. We know that the zero map $0 \in A$ since A is a subalgebra, so this would mean that if $a(v) = f(a)v$ for all $a \in A$, in particular we have $0 = 0(v) = f(0)v = 1v = v$, so the weight space/eigenspace V_f is zero. □

2. Let

$$A = \left\{ \begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix} : a, b \in \mathbb{C} \right\}.$$

(a) Prove that A is an abelian subalgebra of $\mathfrak{gl}(2, \mathbb{C})$.

(b) Let $\lambda : A \to \mathbb{C}$ be defined by $\lambda \begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix} = a+b$. Find the weight space V_λ associated to λ.

Proof.

(a)

$$\begin{aligned} &\left[\begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix}, \begin{pmatrix} c & d \\ 0 & c+d \end{pmatrix} \right] \\ &= \begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix} \begin{pmatrix} c & d \\ 0 & c+d \end{pmatrix} - \begin{pmatrix} c & d \\ 0 & c+d \end{pmatrix} \begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix} \\ &= \begin{pmatrix} ac & ad+bc+bd \\ 0 & ac+ad+bc+bd \end{pmatrix} - \begin{pmatrix} ac & bc+ad+bd \\ 0 & ac+bc+ad+bd \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

(b) Suppose that $v \in V$ and $av = \lambda(a)v$ for all $a \in A$. Then writing $a = \begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix}$,

$$\begin{aligned} \begin{pmatrix} av_1 + bv_1 \\ av_2 + bv_2 \end{pmatrix} &= (a+b)\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \\ &= (a+b)v \\ &= \lambda(a)v \\ &= av \\ &= \begin{pmatrix} a & b \\ 0 & a+b \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \\ &= \begin{pmatrix} av_1 + bv_2 \\ av_2 + bv_2 \end{pmatrix}. \end{aligned}$$

This leaves us with two equations: $av_1 + bv_1 = av_1 + bv_2$ and $av_2 + bv_2 = av_2 + bv_2$. Flearly the second equation is trivial, but the first tells us that $bv_1 = bv_2$. This equality holds if $b = 0$, but we need the equation to hold for all possible values of b, so we are left with $v_1 = v_2$. Thus

$$V_\lambda = \left\{ \begin{pmatrix} x \\ x \end{pmatrix} : x \in \mathbb{C} \right\}.$$

□

3.

(a) Let L be a Lie subalgebra of $\mathfrak{gl}(V)$. Suppose that there is a basis of V such that, with respect to this basis, every $x \in L$ is represented by a strictly upper triangular matrix. Show that L is isomorphic to a Lie subalgebra of $n(n, F)$ and hence that L is nilpotent.

(b) Prove a result analagous to (i) in the case where there is a basis of V such that with respect to this basis every $x \in L$ is represented by an upper triangular matrix.

Proof.

(a) Let $f : L \to n(n, F)$ be defined by taking each $x \in L$ to its strictly upper triangular matrix under the aforementioned basis. $\operatorname{im} f$ is closed under linear combinations since $ax+by \in L$ by virtue of L being a Lie subalgebra, and thus $f(ax+by) = af(x)+bf(y) \in n(n, F) \in \operatorname{im} f$ (which also shows that f is a linear map), and $\operatorname{im} f$ is closed under brackets by similar logic, and likewise with showing that f carries brackets to brackets, showing it is a homomorphism. Of course, f is automatically surjective onto its image, so the only thing left to show is that it is injective. Indeed, if $f(x) = f(y)$, then the matrices for x and y are the same *over the same basis*, so $x = y$, and f is an isomorphism onto $\operatorname{im} f$, which is a subalgebra of $\mathfrak{gl}(V)$.

(b) Nothing in the proof of (a) uses the fact that the matrices were *strictly* upper triangular. The proof is thus identical, modulo the word "strictly".

□

4. Let L be a Lie algebra and let A be a subalgebra of L. The *normalizer* of A, denoted $N_L(A)$, is defined by

$$N_L(A) = \{x \in L : [x, a] \in A \text{ for all } a \in A\}.$$

Prove that $N_L(A)$ is a subalgebra of L and that $N_L(A)$ contains A. Show moreover that $N_L(A)$ is the largest subalgebra of L in which A is an ideal.

Proof. Certainly $A \subseteq N_L(A)$ since A is a subalgebra and thus $[a, b] \in A$ whenever $a, b \in A$. If $x, y \in N_L(A)$ and $\alpha, \beta \in F$, then given $a \in A$,

$$[\alpha x + \beta y, a] = \alpha[x, a] + \beta[y, a] \in A$$

since both $[x, a] \in A$ and $[y, a] \in A$ and A is a subalgebra, so $N_L(A)$ is closed under linear combinations. Furthermore,

$$[[x, y], a] = \underbrace{[\overbrace{[x, a]}^{\in A}, y]}_{\in A} + \underbrace{[\overbrace{[a, y]}^{\in A}, x]}_{\in A} \qquad \text{(by Jacobi)}$$

where every instance of "$\in A$" comes from taking the bracket of an element in $N_L(A)$ with an element of A. Thus $N_L(A)$ is a subalgebra of L.

Now suppose B is a subalgebra of L which contains $N_L(A)$ as a proper subset, and let $b \in B \setminus N_L(A)$. Then there exists an $a \in A$ such that $[b, a] \notin A$ (since otherwise b would be in $N_L(A)$), so A is not an ideal of B. Thus $N_L(A)$ is the largest subalgebra in which A is an ideal. □

5. Let $L = \mathfrak{gl}(n, \mathbb{C})$ and let A be the subalgebra of L consisting of all diagonal matrices. Show that $N_L(A) = A$.

Proof. As the hint says, we can use the invariance lemma here, concluding that since multiplying a diagonal matrix by another matrix has the effect of just multiplying each nth row of the second matrix by the n, n entry in the diagonal matrix, for that resultant matrix to be diagonal, the one on the right must have been diagonal in the first place. That gives us the inclusion $N_L(A) \subseteq A$, and $A \subseteq N_L(A)$ already by prior work, so we have $N_L(A) = A$. □

6. Show that if $a, y \in \mathfrak{gl}(V)$ are linear maps, then for any $m \geq 1$

$$ay^m = y^m a + \sum_{k=1}^{m} \binom{m}{k} y^{m-k} a_k$$

where $a_1 = [a, y]$, $a_2 = [a_1, y] = [[a, y], y]$, and generally $a_k = [a_{k-1}, y]$. Deduce that

$$\operatorname{ad} y^m = \sum_{k=1}^{m} (-1)^{k+1} \binom{m}{k} y^{m-k} (\operatorname{ad} y)^k.$$

In this formula, the power y^{m-k} acts on $\mathfrak{gl}(V)$ as the linear map which sends $x \in \mathfrak{gl}(V)$ to $y^{m-k}x$.